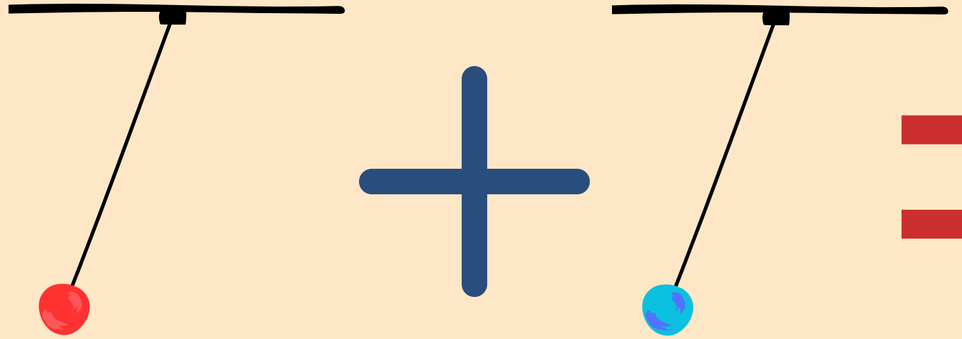
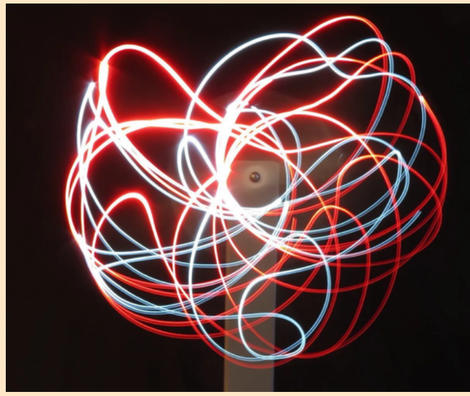
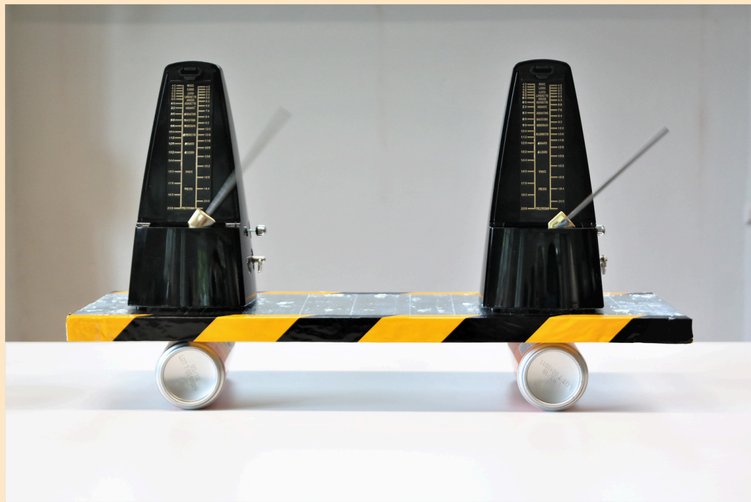






ARythm



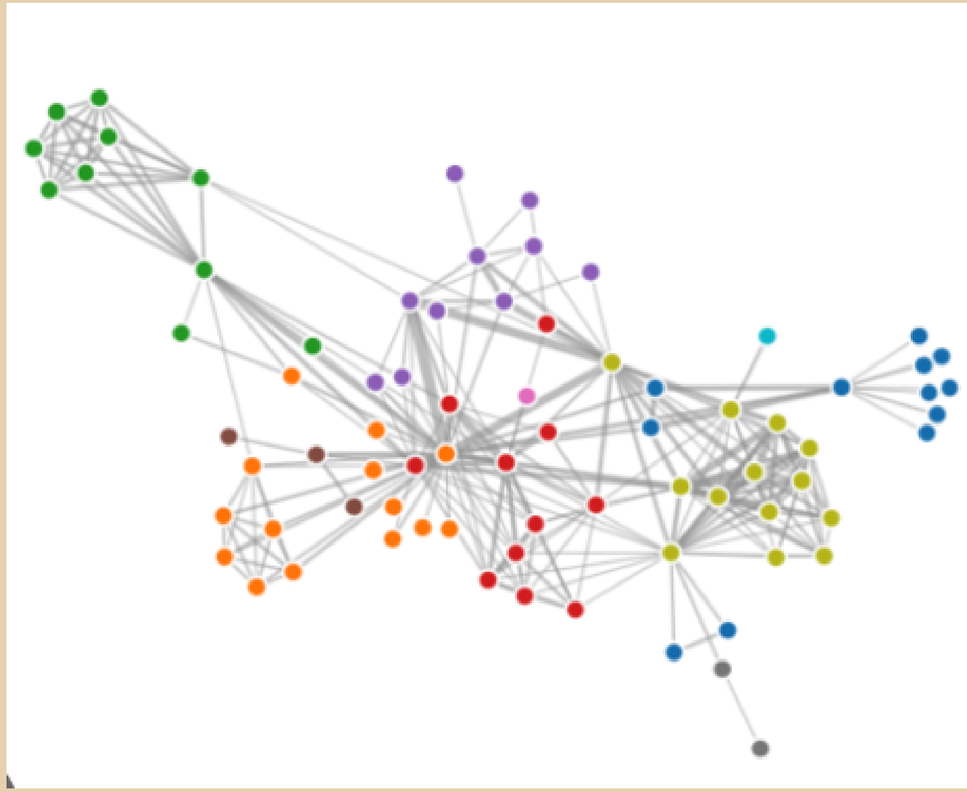


Problem statement

We tested the stability of a network of edges and vertices representing a system which can be modelled by a system of first order linear differential equations.

$$Ax = \dot{x}$$

We can write the system of equations using matrices and vectors.



Condition for stability of the system

The system will be stable if the real part of the eigenvalues are non-positive.

$$e^A = \sum_{i=0}^{\infty} \frac{A^i}{i!}$$

$$e^{At} = \sum_{i=0}^{\infty} \frac{A^i t^i}{i!}$$

(e^{At}) v is a solution to $Ax = \dot{x}$

$$\text{as } \frac{d}{dt}((e^{At})v) = Ae^{At}v$$

$$A = PDP^{-1}$$

$$e^{At} = P\left(\sum_{i=0}^{\infty} \frac{D^i t^i}{i!}\right)P^{-1}$$

$$\begin{bmatrix} e^{\lambda_1 t} & 0 & \cdots & 0 \\ 0 & e^{\lambda_2 t} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{\lambda_n t} \end{bmatrix}$$

- **Initialized a matrix with negative eigenvalues - to ensure stability**

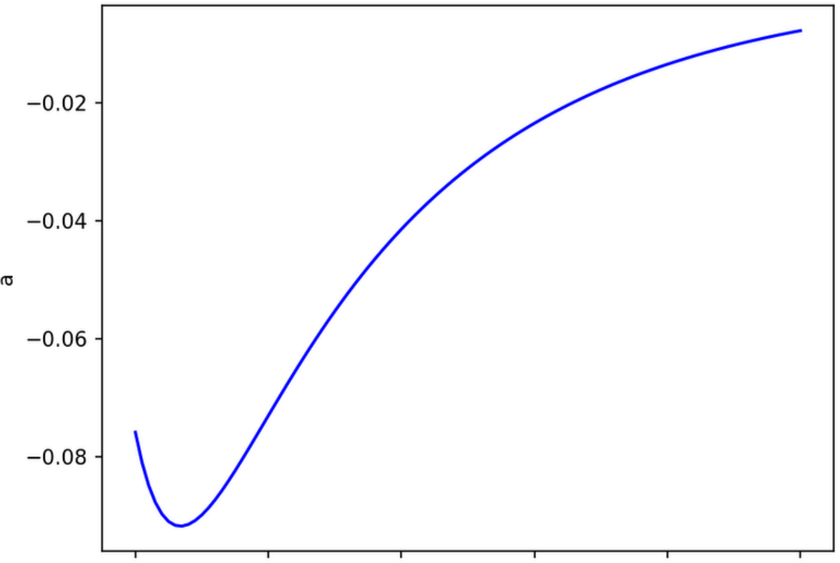
$$\begin{bmatrix} -0.22255878 & 0.09121193 & 0.14205416 & 0 & 0 & 0.02017017 & -0.04770694 \\ 0.09121193 & -0.30613141 & -0.08175898 & 0.03912507 & 0.0329546 & 0.0109875 & -0.02598788 \\ 0.14205416 & -0.08175898 & -0.38096664 & -0.02512192 & -0.0211599 & 0.01711202 & -0.04047372 \\ 0 & 0.03912507 & -0.02512192 & -0.51000072 & 0.14678499 & 0.01482777 & 0.00626908 \\ 0 & 0.0329546 & -0.0211599 & 0.14678499 & -0.56063471 & -0.01760415 & -0.00744291 \\ 0.02017017 & 0.0109875 & 0.01711202 & 0.01482777 & -0.01760415 & -0.6384538 & -0.11181869 \\ -0.04770694 & -0.02598788 & -0.04047372 & 0.00626908 & -0.00744291 & -0.11181869 & -0.42125392 \end{bmatrix}$$

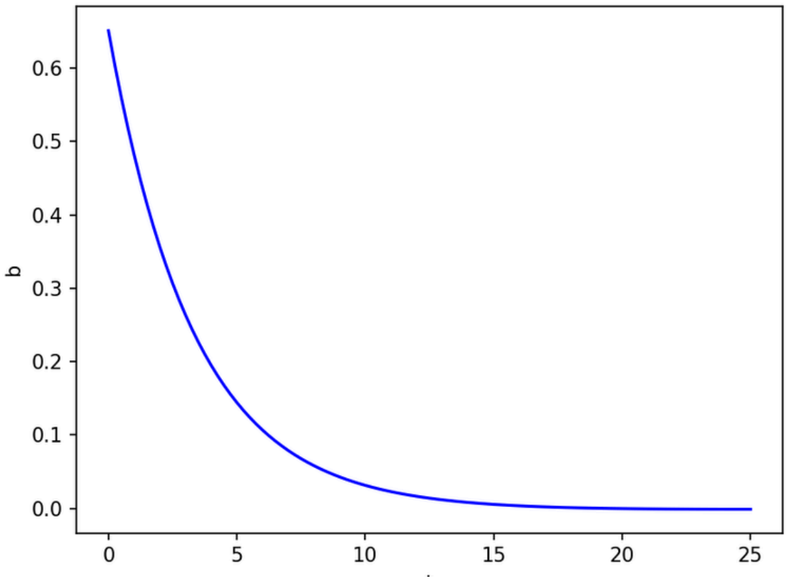
- **to represent the ADJACENCY matrix of the network by arbitrarily adding some zeroes to signal absence of edges in the network.**

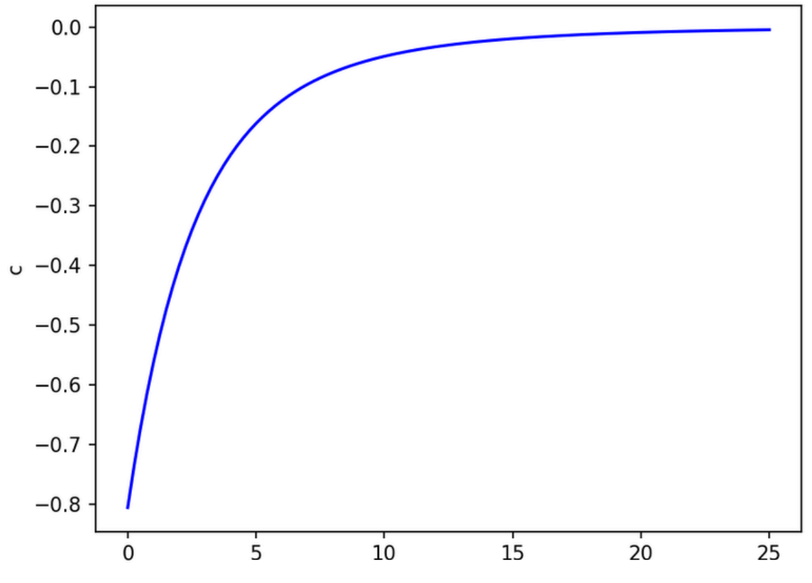
- **We randomly initialized a vector to represent the initial conditions:**

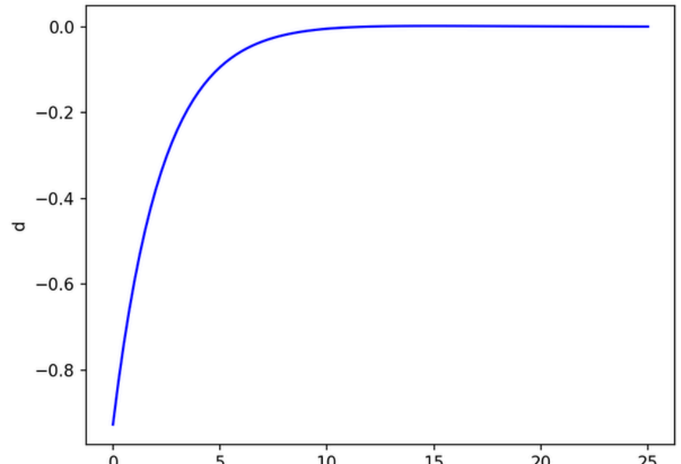
$$\begin{bmatrix} -0.07581242 \\ 0.65113902 \\ -0.8065105 \\ -0.92616897 \\ -0.8194442 \\ 0.98481995 \\ 0.09813605 \end{bmatrix}$$

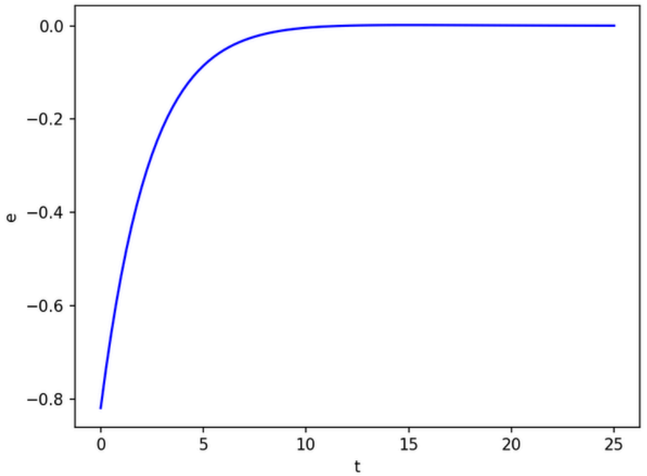
We solved the system numerically to obtain the solution to the system of differential equations

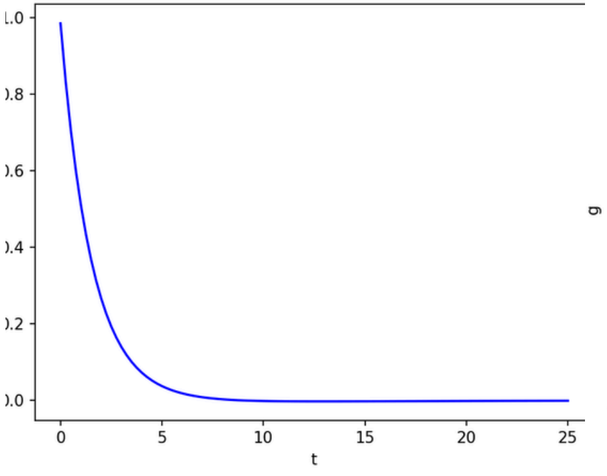


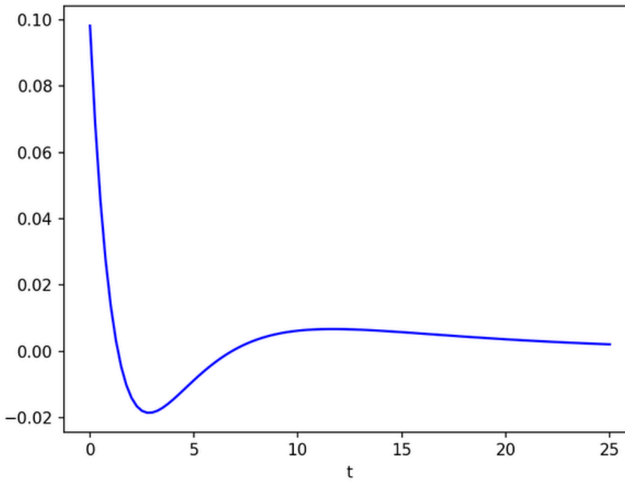












Synchronization

A dynamic process in which some systems tend to synchronize their motion depending on how they are connected.

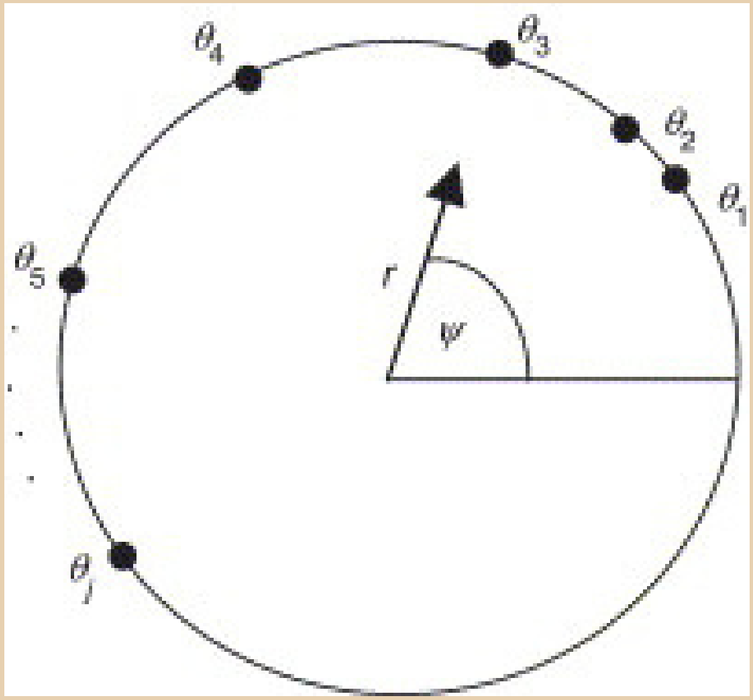


One successful attempt to study synchronization is the Kuramoto model

The Kuramoto model for a population of N oscillators with global all to all coupling is given by

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^{N} \sin(\theta_j - \theta_i)$$

for i=1,2,...,N



Each oscillator has its own natural frequency ω which is drawn randomly from specified distribution g(ω)(specifically a probability density) that is normalized.

complex order parameter r(t) and average phase ψ(t):

$$r(t)e^{i\psi(t)} = \frac{1}{N}\sum_{j=1}^{N} e^{i\theta_j(t)}$$

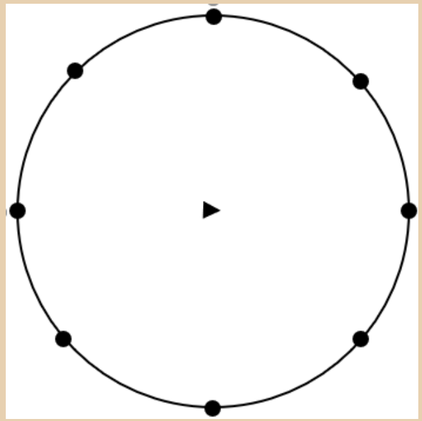
The r(t) gives the measure of the degree of synchrony of the systems of oscillators .

If r(t)=0; complete incoherence.

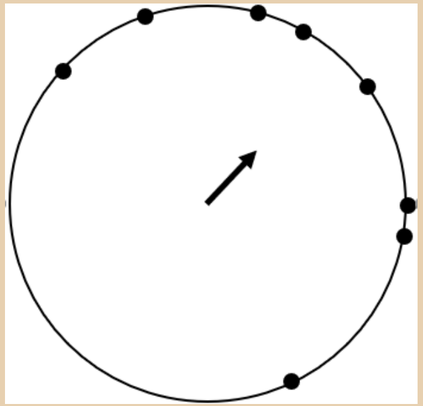
If r(t) = 1;complete synchronisation.

If 0<r(t)<1;partial synchronisation.

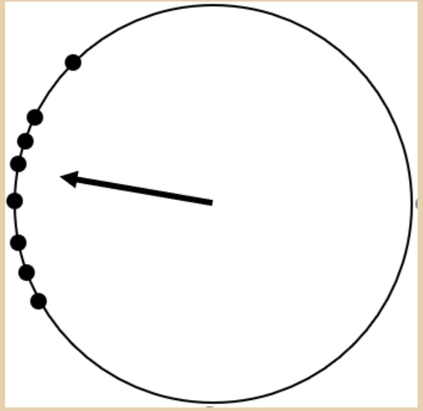
Visualizing the order parameters for N=8 oscillators



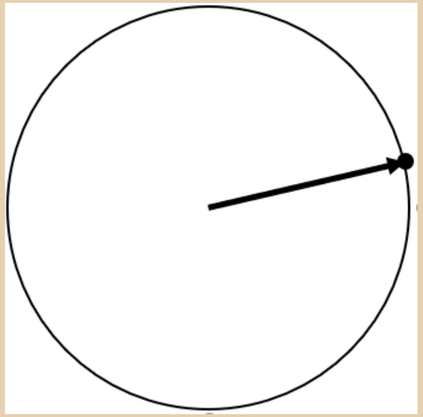
(a) r = 0.0



(b) r = 0.4



(c) r = 0.8



(d) r = 1.0

The 'center of mass' of the phases is represented by the complex number re^iψ and is visualised by the vector of length r and angular displacement ψ

Using the order parameter and the average phase the equation can be rewritten as:

$$\dot{\theta} = \omega_i + Kr\sin(\psi - \theta_i), i = 1, 2, \ldots N$$

This equation describes the system as a mean field model.

If K is too small then no synchronization

If K is too high then synchronization happens

So there will be a critical K where transition happens

Deriving K_c in the continuum limit($N \to \infty$)

We introduce the density function p(θ, ω, t) where p is non-negative,periodic and normalized.

$$\int_0^{2\pi} \rho(\theta, \omega, t) d\theta = 1$$

The conservation of oscillators is expressed as

$$\frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial \theta}(\rho v)$$

This equation describes the evolution of p over time

Now lets play with equations:

$$v(\theta, \omega, t) = \dot{\theta} = \omega + Kr(t)sin\left(\psi - \theta\right)$$

The order parameter is now expressed as:

$$r(t)e^{i\psi(t)} = \int_0^{2\pi} \int_{-\infty}^{\infty} e^{i\theta} \rho(\theta, \omega, t) g(\omega) d\omega d\theta$$

The final equation now becomes :

$$\frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial \theta}\left[\rho\left(\omega + K\int_0^{2\pi} \int_{-\infty}^{\infty} \sin(\theta' - \theta)\rho(\theta', \omega', t) g(\omega')\, d\omega' d\theta'\right)\right]$$

Now the before equations has a trivial solution:

$$\rho(\theta, \omega) = \frac{1}{2\pi}, r = 0$$

This corresponds to the complete incoherent state of the oscillators.

Going into a frame rortating with frequency Ω and applying steady state conditions and WLOG taking ψ = 0 we get

$$\dot{\theta} = \omega_i - Kr\sin(\theta_i)$$

$$\rho(\theta, \omega) = \frac{C(\omega)}{|\dot{\theta}|} = \frac{C(\omega)}{|\omega - Kr\sin\theta|}$$

$$\int_{-\pi}^{\pi} \rho(\theta, \omega)\, d\theta = C(\omega) \int_{-\pi}^{\pi} \frac{d\theta}{|\omega - Kr\sin\theta|} = 1$$

we will have two types of oscillators locked and drifting oscillators and their density function

$$\rho(\theta,\omega) = \begin{cases} \delta(\theta - \arcsin(\frac{\omega}{Kr})), & |\omega| \le Kr, \quad locked \\ \frac{\sqrt{\omega^2-(Kr)^2}}{2\pi|\omega - Kr\sin\theta|}, & |\omega| > Kr, \quad drifting \end{cases}$$

Now solving for r in terms of K using above functions

$$r(t)e^{i\psi(t)} = \langle e^{i\theta} \rangle = \langle e^{i\theta} \rangle_{locked} + \langle e^{i\theta} \rangle_{drifting}$$

$$\langle e^{i\theta} \rangle = re^{i\psi} = r \qquad \text{(as ψ = 0)}$$

$$r = \langle e^{i\theta} \rangle_{locked} + \langle e^{i\theta} \rangle_{drifting}$$

$$\langle e^{i\theta} \rangle_{drifting} = \int_0^{2\pi} \int_{|\omega|>Kr} \rho(\theta,\omega) g(\omega) e^{i\theta} d\omega d\theta$$

$$\langle e^{i\theta} \rangle_{drifting} = \int_0^{2\pi} \int_{-\infty}^{-Kr} \rho(\theta,\omega) g(\omega) e^{i\theta} d\omega d\theta + \int_0^{2\pi} \int_{Kr}^{\infty} \rho(\theta,\omega) g(\omega) e^{i\theta} d\omega d\theta$$

$$\langle e^{i\theta} \rangle_{drifting} = -\int_{-\pi}^{\pi} \int_{Kr}^{\infty} \rho(\theta'+\pi,-\omega) g(-\omega) e^{i\theta'}\, d\omega\, d\theta' + \int_0^{2\pi} \int_{Kr}^{\infty} \rho(\theta,\omega) g(\omega) e^{i\theta}\, d\omega\, d\theta$$

$$\langle e^{i\theta} \rangle_{\text{drifting}} = -\int_{-\pi}^{\pi} \int_{Kr}^{\infty} \rho(\theta',\omega) g(\omega) e^{i\theta'}\, d\omega\, d\theta' + \int_0^{2\pi} \int_{Kr}^{\infty} \rho(\theta,\omega) g(\omega) e^{i\theta}\, d\omega\, d\theta$$

$$\langle e^{i\theta} \rangle_{\text{drifting}} = -\int_0^{2\pi} \int_{Kr}^{\infty} \rho(\theta',\omega) g(\omega) e^{i\theta'}\, d\omega\, d\theta' + \int_0^{2\pi} \int_{Kr}^{\infty} \rho(\theta,\omega) g(\omega) e^{i\theta}\, d\omega\, d\theta$$

$$\langle e^{i\theta} \rangle_{drifting} = 0$$

$$\langle e^{i\theta} \rangle_{locked} = \int_0^{2\pi} \int_{|\omega| \leq Kr} \rho(\theta, \omega) g(\omega) e^{i\theta} \, d\omega \, d\theta$$

$$\langle e^{i\theta} \rangle_{locked} = \int_{-Kr}^{Kr} g(\omega) \left(\cos(\theta(\omega)) + i \sin(\theta(\omega)) \right) d\omega$$

$$\langle e^{i\theta} \rangle_{locked} = \int_{-Kr}^{Kr} g(\omega) \cos(\theta(\omega)) \, d\omega)$$

$$r = \langle e^{i\theta} \rangle_{locked} + \langle e^{i\theta} \rangle_{drifting}$$

$$r = Kr \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} g(Kr \sin\theta) \cos^2\theta \, d\theta$$

r=0 is a solution to the above equation

$$1 = K \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} g(Kr \sin\theta) \cos^2\theta \, d\theta$$

This solution only exists for certain values of K and bifurcates from the r = 0 branch at some critical value Kc. To find this critical value, we let $r \to 0+$

$$1 = K_c \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} g(0) \cos^2\theta \, d\theta$$

$$K_c = \frac{2}{\pi g(0)}$$

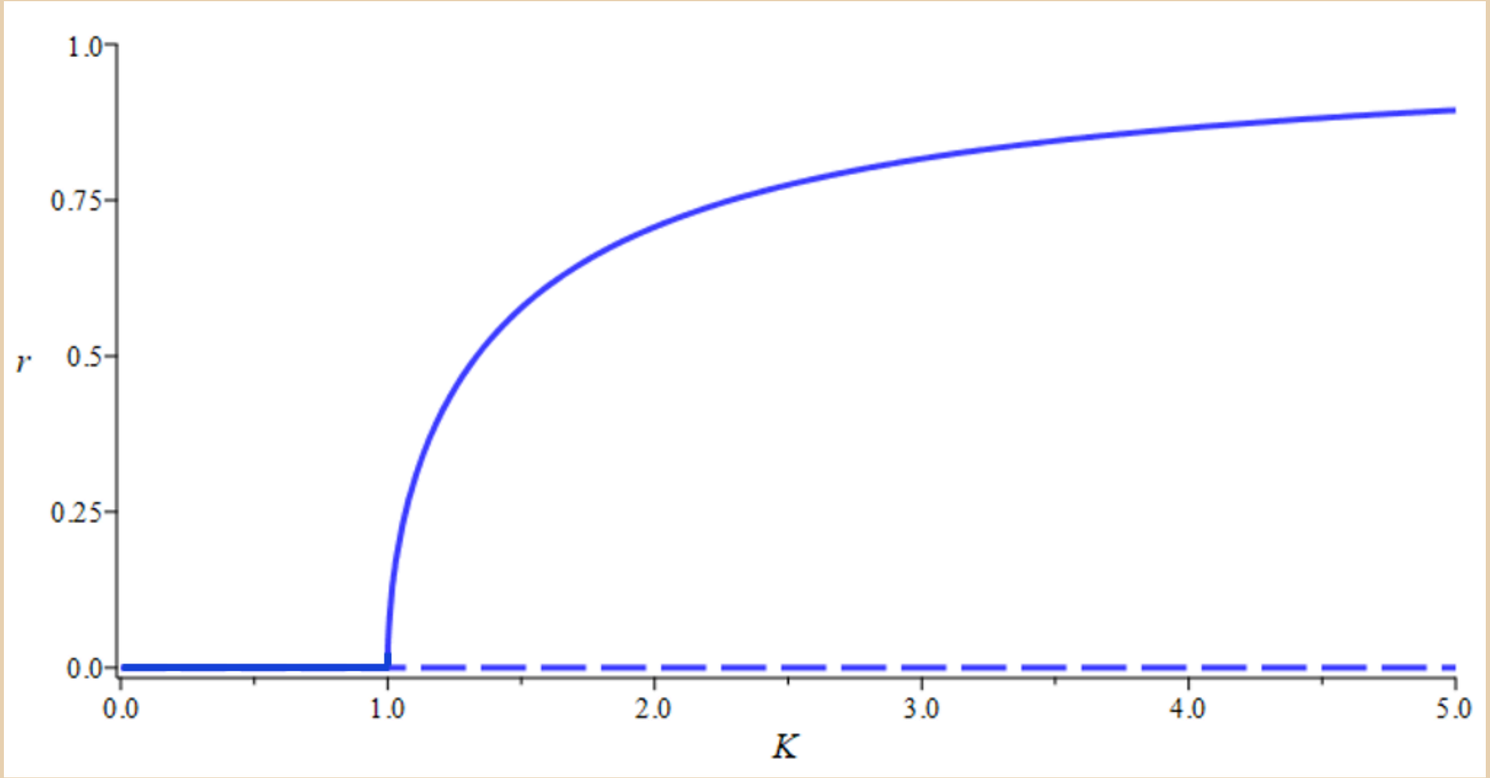
So the above value of K is where the system starts to form clusters or synchronize

If we take g as lorentzian distribution the integral is perfectly solvable and we get

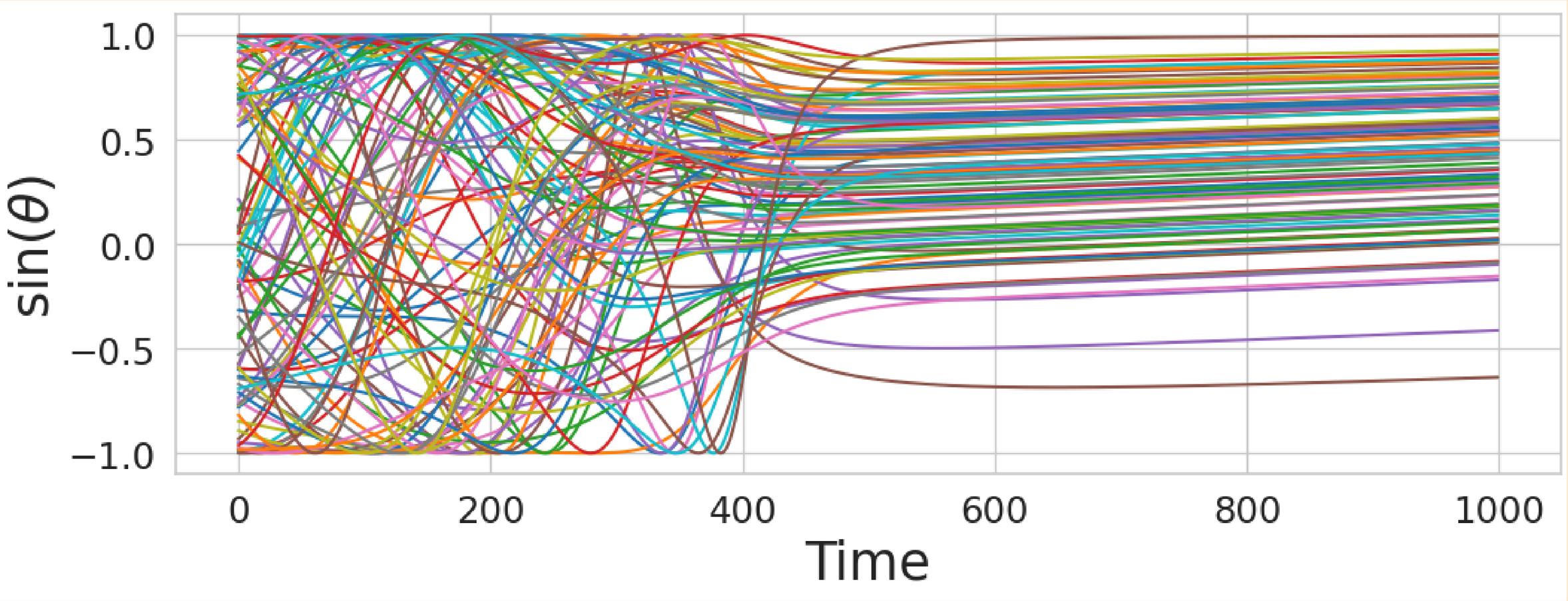
$$r = \sqrt{1 - \frac{K_c}{K}}$$

Bifurcation Analysis

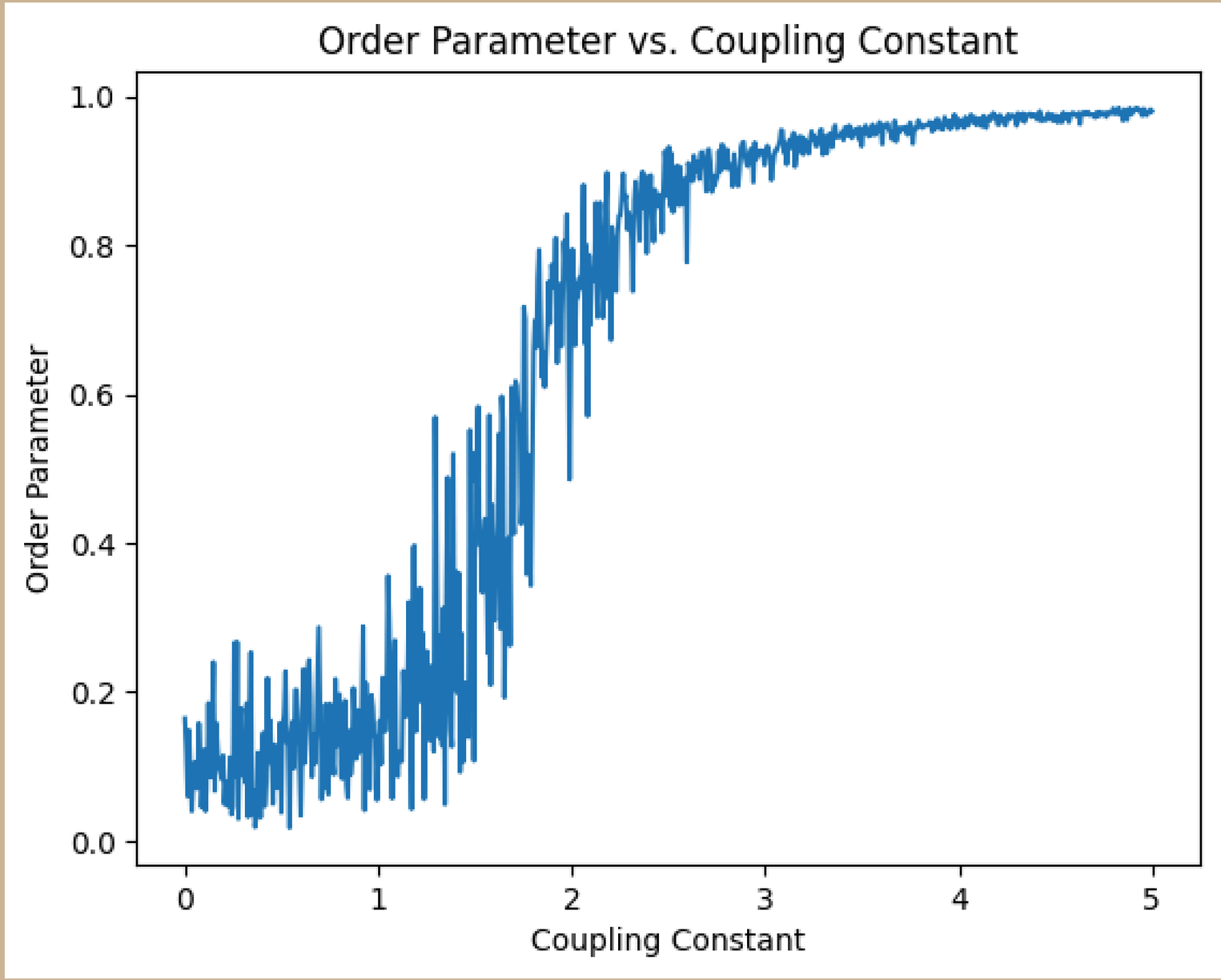
As at critical point the value of r transitions from one stable equilibrium point to two stable and one unstable equilibrium point so this is a supercritical pitchfork bifurcation.

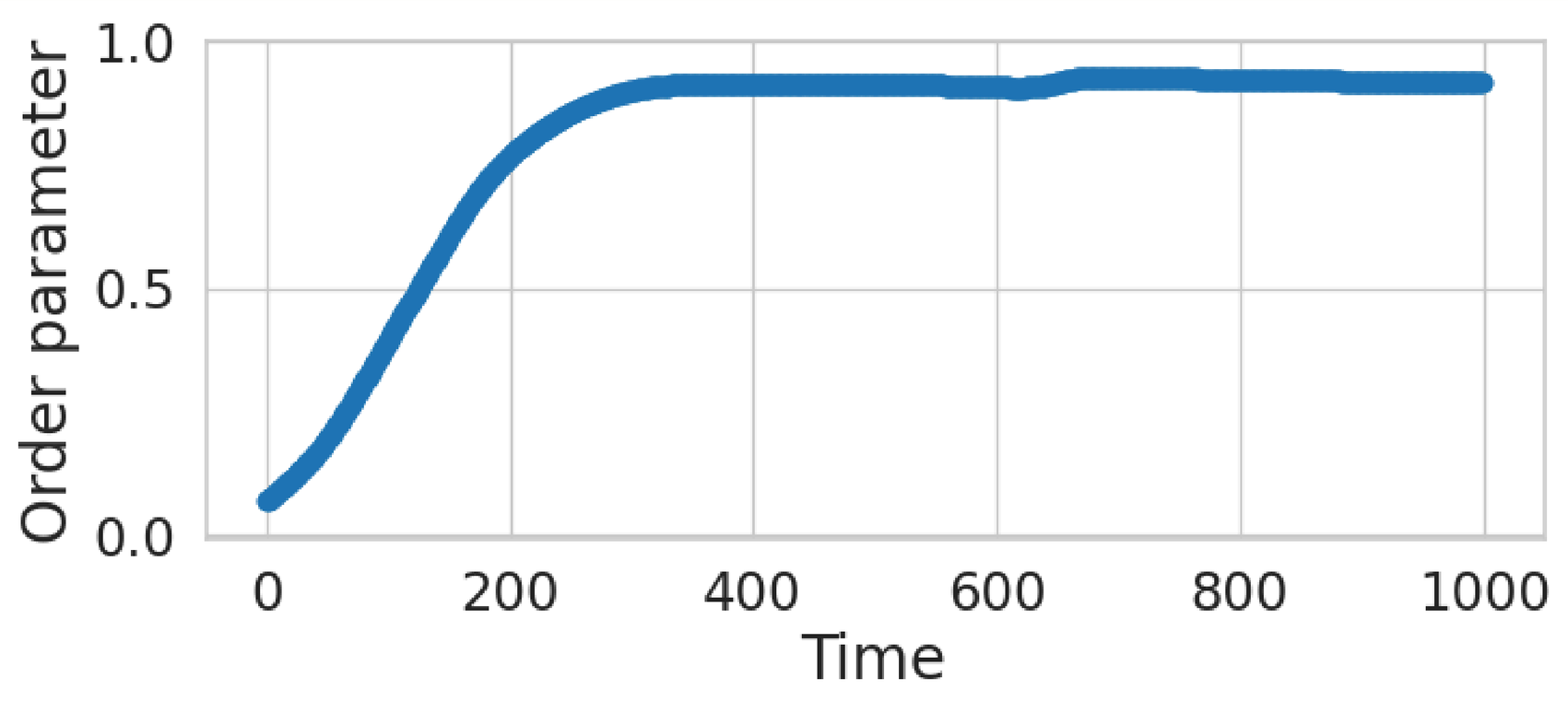


Simulation

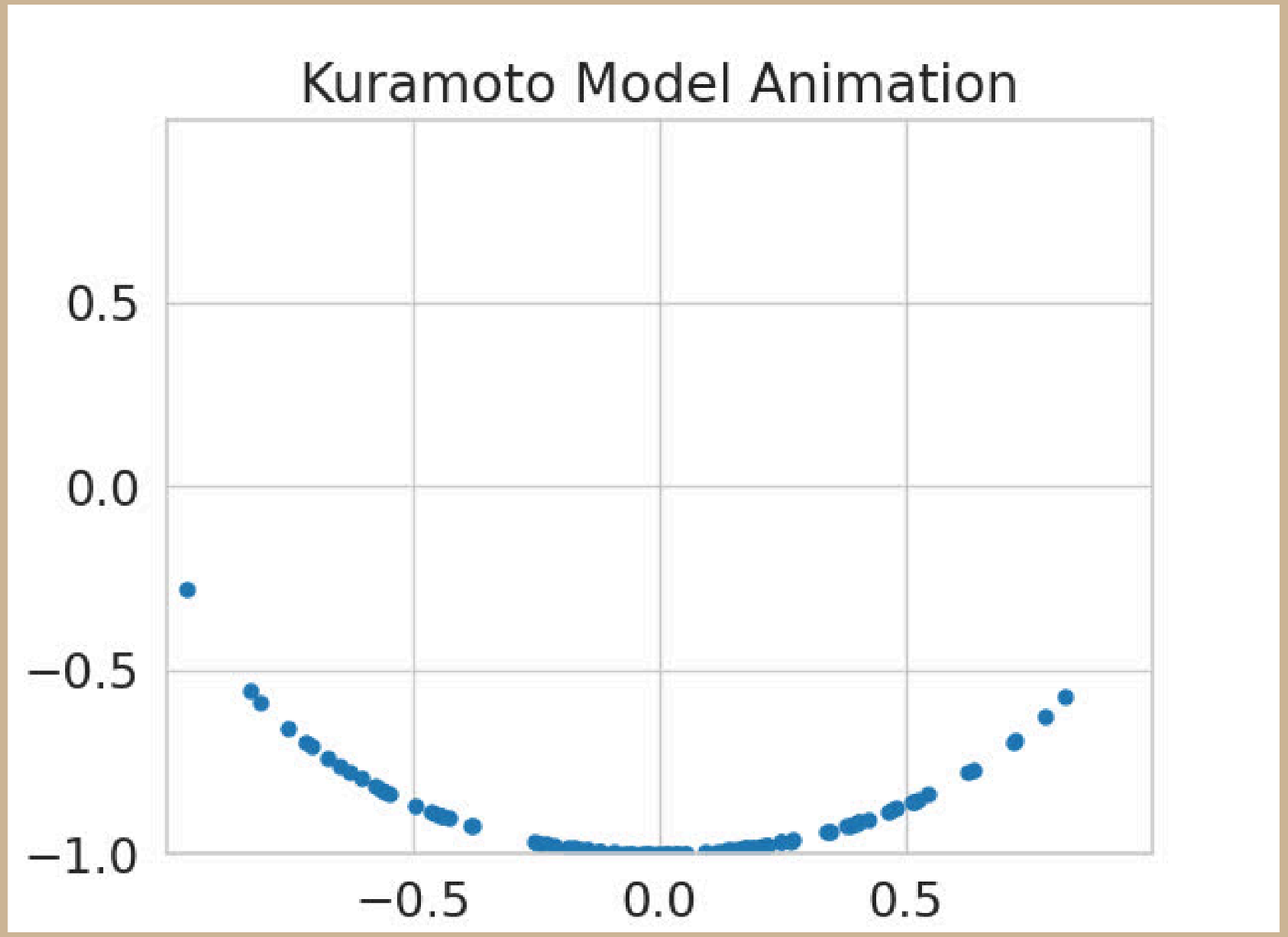


Initializing 100 oscillators with their natural frequencies in a normal distribution and their phases uniformly distributed


Order Parameter vs. Coupling Constant
Order Parameter
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
Coupling Constant



Order parameter v/s time (coupling constant is greater than critical value)


Kuramoto Model Animation
0.5
0.0
−0.5
−1.0
−0.5
0.0
0.5





Thank you